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Pollution and innovation: Evidence from manufacturing firms in China

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Abstract
By constructing a comprehensive database containing firm-level and meteorological-level information, this study examines how air pollution affects firm innovation. Using an instrumental variable approach, we find that air pollution significantly negatively impacts firm innovation. For every increase of 1 μg/m^3 in the $PM_{2.5}$ concentration, the number of patent applications will be significantly reduced by 0.72%. In addition, the concentration of $PM_{2.5}$ is detrimental to the quality of patents. We also find that the negative effect of air pollution on firm innovation is more pronounced for private firms and capital-intensive firms. Mechanistic tests show that air pollution hinders the accumulation of innovative workers and increases production costs. Our findings indicate that firms' behavioral responses should be accounted for when assessing how air pollution affects social welfare.

Keywords Air pollution · Firm innovation · Thermal inversion

1 Introduction

Innovation has been recognized as the main engine for economic growth (Solow 1957; Romer 1990; Aghion and Howitt 1992). The introduction of new products or new processes can significantly improve the production efficiency of firms. Compared with an increase in capital or labor, improving efficiency has a greater impact

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on economic growth. Most studies on developed countries have found that innovation is the primary source of economic growth (Ulku 2007). For developing countries, such as China, studies have also found that productivity improvement is one of the crucial reasons for China's rapid economic growth (Chow and Lin 2002; Bosworth and Collins 2008). China's investment in research and development (R&D) has sustained robust growth over a prolonged period. In 2018, China's R&D investment accounted for 2.19% of its GDP, surpassing the EU-15 average (2.13%). China's number of invention patent applications in 2018 reached 1.542 million, accounting for approximately 46.4% of the global total[1]. Although China's R&D investment and the number of patent applications have proliferated, the proportion of firms with patent application activities is still low. Kou & Liu (2020) found that the proportion of Chinese industrial firms applying for patents is 11.4%. If only invention patents are considered, this proportion will further drop to 6.6%.

The literature on firm innovation either focuses on the impact of microlevel characteristics, such as financing constraints and capital structure (Hall and Lerner 2010; Ayyagari et al. 2011), or analyzes the impact of macro policies, such as tax incentives, fiscal subsidies, and intellectual property protection, on innovation (Bloom et al. 2002; Ernst et al. 2014; Dechezlepretre et al. 2016; Akcigit et al. 2018; Chen et al. 2018a, b). This study sheds light on the impact of air pollution on firm innovation. Although China's air quality has improved in recent years, pollution management still has a long way. According to a report released by the Ministry of Ecology and Environment in 2019, China's average concentration of $PM_{2.5}$ in 337 prefecture-level cities is 36 μg/m^3, which is higher than the safety value of 10 μg/m^3 given by the World Health Organization. 46.6% of all cities meet the standard for ambient air quality[2].

Theoretically, the impact of air pollution on firm innovation is not unidirectional but rather a complex process involving the interplay of positive and negative effects, with the ultimate impact depending on which effect dominates. Specifically, its transmission mechanisms can be analyzed through the following two opposing perspectives.

On one hand, air pollution inhibits firm innovation through mechanisms such as affecting human capital and increasing operating costs. Severe air pollution harms employee health, leading to increased absenteeism and decreased work efficiency, directly undermining the core human capital essential for firm innovation. Furthermore, a poor environment may also influence the location choices of highly skilled talent. In addition, to cope with pollution, firms may need to increase spending on health protection, environmental cleaning, etc. Additionally, potential government environmental penalties or temporary policies like production restrictions or suspensions triggered by pollution could divert innovation resources originally intended for R&D, increasing operational uncertainty and thus creating a "crowding-out effect" on innovation investment.

[1] Please see " World Intellectual Property Indicators in 2019" released by World Intellectual Property Organization (WIPO).

[2] See the website of the Ministry of Ecology and Environment of the People's Republic of China: http://www.mee.gov.cn/.

On the other hand, air pollution may also promote firm innovation through the "induced innovation effect". According to the Porter Hypothesis (Porter and van der Linde 1995), properly designed environmental regulations can stimulate innovation offset within firms, where the benefits can partially offset the compliance costs. Severe air pollution is often a direct trigger for governments to introduce stricter environmental policies. In this context, firms are compelled to seek more efficient and cleaner production technologies to meet new environmental standards. This process forces firms to re-optimize resource allocation and focus innovation efforts on energy saving, emission reduction, and green technologies, thereby potentially inducing enhanced innovation capabilities that improve long-term competitiveness.

In conclusion, the net effect of air pollution on firm innovation is an proposition that requires empirical testing. Research based on the Chinese context is particularly important, as China is in a critical period of economic transformation and environmental pollution control, where the intensity of government environmental regulations and firm response strategies possess unique characteristics.

In this study, we constructed a comprehensive micro database containing firm-level information, innovation information, pollution information, and meteorological information to identify the causal effects of air pollution on firm innovation. However, there are two main challenges in empirical analysis: indicator measurement and endogeneity issues. Regarding the selection of air pollution indicators, China mainly used the air pollution index (API) before 2012, but it includes only three pollutants—PM_{10}, SO_2, and NO_2—not including the key variable $PM_{2.5}$. Moreover, the API indicators cover only some major cities and are not reported at the county level[3]. Although the new air quality index (AQI) reports $PM_{2.5}$, we cannot obtain this information before 2012. Therefore, following Buchard et al. (2016), we use satellite data to retrieve air pollution indicators. First, we identify the monthly pollution grid data at the county level and then calculate the annual average to obtain the concentration of $PM_{2.5}$ for an extended period (1998 to 2013). In addition, for the endogenous bias, we calculated the annual number of days in which the thermal inversion phenomenon occurred at the county level and used it as an instrumental variable for the concentration of $PM_{2.5}$.

We found that air pollution has a significant negative impact on firm innovation once endogeneity is considered. For every increase of 1 $\mu g/m^3$ in the $PM_{2.5}$ concentration in the county where the firm is located, this firm's number of patent applications will be significantly reduced by 0.72%. This conclusion is maintained after a series of robustness tests, such as those considering environmental regulations. In addition, the concentration of $PM_{2.5}$ is detrimental to the quality of patents. Heterogeneity analysis revealed that the concentration of $PM_{2.5}$ has a more significant impact on private and capital-intensive firms' innovation. Further research on the mechanism revealed that the concentration of $PM_{2.5}$ is not conducive to the accumulation of human capital and increases the production costs of firms.

The remainder of this paper is organized as follows. Section 2 introduces the related literature. Section 3 introduces the background of air pollution in China.

[3] The API indicators were reported only after June 2000, and it was initially announced in 42 major cities, and it increased to 86 cities in 2010.

Section 4 presents the empirical methodologies and describes the data. Section 5 reports the empirical findings. Finally, the conclusions of this study are summarized in Section 6.

2 Literature review

This article is closely related to two streams of literature. First, it is related to the literature that analyzes the impact of air pollution. A large body of literature focuses on the impact of air pollution on individual health and cognitive ability. Long-term exposure to a polluted environment significantly increases the probability of individuals suffering from respiratory diseases, immune diseases, cardiovascular diseases, cerebrovascular diseases, and lung cancer (Brunekreef and Holgate 2002; Künzli et al. 2005). Research in China has also revealed that air pollution is not conducive to public health, and in the long run, it will also negatively affect the life expectancy of residents (Chen et al. 2013; Ebenstein et al. 2015). Air pollution can also significantly change individuals' cognitive abilities and induce unhealthy lifestyle habits. Deschenes et al. (2020) reported that air pollution can lead to obesity. Mechanistic analysis revealed that in a polluted environment, individuals decrease physical exercise, have fewer walking choices, sleep less, and have an unhealthy diet. Herrnstadt et al. (2020) further found that pollution changes personal health and cognitive ability and triggers violent behavior.

In addition to affecting individuals, air pollution can also significantly affect firm behavior. Fu et al. (2021) found that air pollution significantly reduces firms' labor productivity. Xue et al. (2021) found that air pollution is not conducive to the accumulation of high-skilled talent in listed firms. Several recent papers are also concerned with the impact of air pollution on firm innovation (Xue et al. 2021; Tan and Yan 2021).

Our paper is also related to the extensive literature on firm innovation. Due to the externalities of innovation, most countries worldwide have introduced a series of policies to promote innovation in firms, including tax incentives, accelerated depreciation, investment credits, and financial subsidies. Many studies believe that these policies significantly impact firm innovation (Hall and Van Reenen 2000; Bloom et al. 2002; Dechezlepretre et al. 2016; Akcigit et al. 2018; Guceri and Liu 2019). Research in China has also shown similar results (Howell 2016; Chen et al. 2018a, b). External institutional factors such as intellectual property protection also have a significant impact on firm innovation. Establishing a strict property rights protection system can reduce the risk of innovation knowledge being imitated, enhance the exclusiveness of innovative technology, and ensure that firms maximize their R&D income and increase their enthusiasm for R&D activities (Keller 2004; Cohen 2010). Another branch of literature focuses on the effect of corporate financial characteristics, such as financing constraints, cash flow, and financial leverage, on R&D investment (Hall and Lerner 2010; Ayyagari et al.2011). We contribute to this literature by showing that air pollution is also an important determinant of firm innovation.

Compared with existing research, the main contributions of this study are reflected in the following three aspects:

First, in terms of research objects and data, this paper takes Chinese industrial firms as an example to explore the impact of air pollution on firm innovation. On the one hand, China has experienced severe air pollution and a rapid industrialization process, thus the research conclusions will provide insights for other developing countries. On the other hand, the application of the industrial firms expands the scope of research samples. Compared with listed companies, industrial firms are more representative. In addition, different from ground monitoring data such as the AQI, we use satellite data to calculate the concentration of PM2.5. Satellite data enable our research scope to be accurately defined at the county level.

Second, in terms of research methods, to identify the causal relationship between air pollution and firm innovation, we mainly use the instrumental variable method. In addition, we further provide results obtained based on the regression discontinuity strategy. The use of multiple identification strategies makes our research results have more accurate causal inference.

Finally, we also discover the differential impacts and transmission mechanisms of air pollution on firm innovation. The inhibitory effect of pollution on firm innovation is more evident in samples of private firms and capital-intensive firms, and it affects firm innovation through mechanisms such as undermining human capital accumulation and increasing production costs, which is conducive to a more comprehensive assessment of the impact of air pollution on firm innovation.

3 Background

After the reform and opening-up, China's economy achieved remarkable growth. In 2019, China's per capita GDP exceeded USD10 000, reaching a middle-to-high income level. However, part of the price of rapid economic growth is severe environmental costs. The large consumption of fossil fuels such as coal and relatively loose environmental supervision have seriously affected China's environmental quality. Figure 1 shows the trend of the average concentration of $PM_{2.5}$ in China from 1980 to 2019 based on satellite data. The concentration of $PM_{2.5}$ was low before 2000, but it showed a remarkable upward trend from 2000 to 2013. This rapid growth trend was effectively curbed after 2014; in the following years, it showed a downward trend. As of 2019, although the average concentration of $PM_{2.5}$ in China was still significantly higher than the safety line given by the World Health Organization (10 $\mu g/m^3$), the concentration of $PM_{2.5}$ decreased by 15% compared to that in 2010.

Poor air quality can cause a series of diseases, such as respiratory diseases and cardiovascular and cerebrovascular diseases, which are not conducive to the public health of residents (Brunekreef and Holgate 2002; Künzli et al. 2005). For example, lung cancer is closely related to air quality. According to the statistics of the National Cancer Center of China, the incidence and death of lung cancer have ranked first among malignant tumors for many years. In 2015, approximately 631,000 deaths occurred due to lung cancer, and the mortality rate was 45.87 per 100,000 people. A reduction in the public health of residents is not conducive to the long-term development of the country.

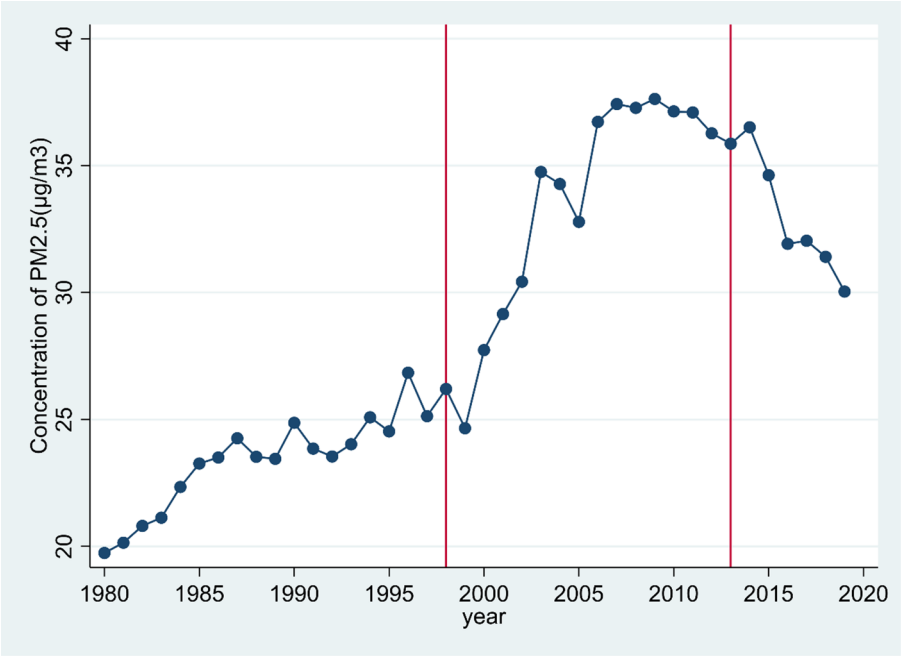

Fig. 1 Time Trend of the Concentration of $PM_{2.5}$ in Mainland China (1980–2019).
Notes: This figure depicts the national average $PM_{2.5}$ concentration in each year from 1980 to 2019. Two vertical lines highlight our study period: 1998–2013

However, after 2013, the Chinese government increased its environmental governance efforts. At the opening of the annual meeting of the People's Congress in 2014, Premier Li Keqiang declared a "war on pollution". He further noted that the government should make serious efforts to strengthen pollution prevention and control. Focusing on megacities and regions with frequent smog, with delicate particulate matter ($PM_{2.5}$) and inhalable particulate matter (PM_{10}) as breakthroughs, a nationwide real-time monitoring and disclosure system for air quality should be established.

4 Empirical strategy and data

4.1 Empirical strategy

To identify the effect of air pollution on firm innovation, we estimate the following model.

$$Y_{ijt} = \alpha_0 + \alpha_1 P_{jt} + \alpha_2 X_{it} + \alpha_3 W_{jt} + \mu_i + \mu_t + \epsilon_{it} \tag{1}$$

where subscripts i, j, and t represent the firm, county and year, respectively. Y_{ijt} is defined as the logarithm of the number of patent applications, which measures the level of innovation for firm *i* located in county *j* during year *t*. In addition, a patent can

be further divided into invention patents, utility model patents, and design patents. We use a total of three patents to measure innovation levels.

We use P_{jt} to denote the average concentration of $PM_{2.5}$ in county *j* during year *t*. Thus, α_1 is the coefficient that we focus on, which captures the effect of air pollution on firm innovation. X_{it} contains a set of firm-level covariates, which includes the firm's size, leverage, age, capital density, inventory density, and profitability. Since we are concerned about the impact of air pollution on firm innovation, it is necessary to control for other weather variables related to air pollution. We aggregate the annual weather variables W_{jt}, such as precipitation, air pressure, wind speed, average temperature, water pressure, humidity, and sunshine hours, to the county level to eliminate meteorological factors (Arceo et al. 2016). μ_i is the firm fixed effect, μ_t is the year fixed effect, and ϵ_{it} is the error term. Following Fu et al. (2021), we cluster the standard errors (SEs) at the firm level in our baseline estimation, but we also show robustness to other clustering patterns.

The main challenge of identifying the causal impact of air pollution on firm innovation is endogeneity bias. Some omitted variables may both affect air pollution and firm innovation. For example, air pollution may be related to economic development. On the one hand, with a high concentration of industrial activities, high pollution may occur. However, agglomeration also provides inherent advantages for technology diffusion, which promotes firm innovation. In addition, although the concentration of $PM_{2.5}$ changes at the county level and the dependent variable of concern is firm-level variables, reverse causality may still exist. For example, firms with high innovation levels may reduce pollution emissions due to the use of advanced technology or equipment, which in turn reduces the level of air pollution at the regional level.

This study uses an instrumental variable approach to address endogeneity bias. A valid instrumental variable needs to be highly correlated with endogenous variables and at the same time have sufficient exogeneity. We use meteorological conditions to determine the IV for air pollution, particularly the annual number of days with at least one thermal inversion for each county. Usually, the atmospheric temperature decreases with increasing altitude. That is, the atmospheric temperature of the higher layer is always lower than that of the lower layer. In this case, the stratification of the atmosphere is prone to convection, which can transport air pollutants near the ground to high altitudes or other areas, reducing the degree of pollution. However, in some exceptional cases, the atmospheric temperature increases with increasing altitude, which causes the atmospheric temperature of the higher layer to be higher than that of the lower layer. Meteorologists generally call this situation *thermal inversion*[4].

When there is a thermal inversion, the air pollutants near the ground do not spread to the upper air, staying in the same place for a long time, resulting in severe pollution (Malek et al. 2006; Bailey et al. 2011). Following Arceo et al. (2016), we calculate thermal inversions using the difference between the first and second layers (110 and 330 m, respectively). Figure 2 below shows the scatter plot and regression line of the average concentration of $PM_{2.5}$ and the average number of days with at least one thermal inversion at the county level from 1998 to 2013. The number of days of thermal inversion and the concentration of $PM_{2.5}$ are significantly positively correlated.

[4] There are five kinds of situations for the formation of thermal inversion: radiation, advection, topography, subsidence, and ocean current, all of them belong to climate phenomenon.

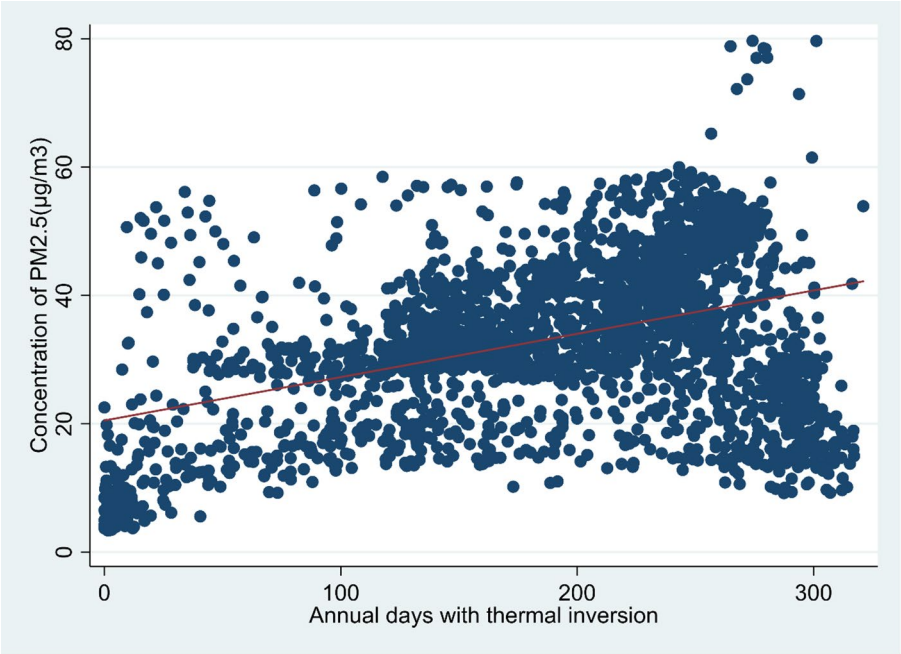

Fig. 2 Average concentration of PM2.5 and annual days with thermal inversion at the county level in mainland China (1998–2013)

Since thermal inversion is a natural phenomenon, theoretically, it will not be related to local economic development (Arceo et al. 2016; Chen et al. 2018a, b; Deschenes et al. 2020; Fu et al. 2021). Thus, we employ the two-stage least squares (2SLS) model as follows:

$$P_{jt} = \beta_0 + \beta_1 TI_{jt} + \beta_2 X_{it} + \beta_3 W_{jt} + \mu_i + \mu_t + \epsilon_{it} \tag{2}$$

$$Y_{ijt} = \gamma_0 + \gamma_1 \widehat{P_{jt}} + \gamma_2 X_{it} + \gamma_3 W_{jt} + \mu_i + \mu_t + \epsilon_{it} \tag{3}$$

where TI_{jt} is the annual days of thermal inversion of county *j* during year *t*. The remaining variables are consistent with Model (1).

4.2 Data

1) *Pollution information*: The concentration of $PM_{2.5}$ is calculated based on *aerosol optical depth (AOD) data* observed by satellites. The *AOD* measures the amount of sunlight absorbed, reflected, and scattered by aerosols and can be used to estimate the concentration of specific substances[5]. An increasing number of studies have

[5] The *AOD* data we use comes from the M2TMNXAER version 5.12.4 product provided by the MERRA-2 (Modern-Era Retrospective analysis for Research and Applications, Version 2) reanalysis data of the

used *AOD* data to retrieve air pollution indicators in recent years. Air pollution indicators calculated based on *AOD* data are highly consistent with indicators measured by ground observation stations and can accurately measure air pollution levels (Gupta et al. 2006; van Donkelaar et al. 2010). We calculated the concentration of $PM_{2.5}$ according to the method provided by Buchard et al. (2016), identified the monthly pollution grid data at the county level and calculated the annual average.

2) *Thermal inversion information*: Thermal inversion data come from the M2I6NPANA version 5.12.4 product provided by the MERRA-2 reanalysis data. Following Arceo et al. (2016), we extracted raster data from counties and calculated the thermal inversion indicator. We calculated the number of days a county experienced thermal inversion each year.
3) *Weather information*: The meteorological data come from the monthly climate database of the *China Meteorological Data Network*, which records detailed information about meteorological indicators, such as temperature, sunshine duration, relative humidity, and wind speed. We extracted weather variables at the county level and calculated the annual average.
4) *Firm data*: The firm-level data are from *the Annual Surveys of Manufacturing Firms (ASMF)* (1998–2013) conducted by China's National Bureau of Statistics (NBS)[6]. It includes annual survey data of all "above scale" industrial firms with more than 5 million RMB in annual sales and contains detailed information on firm characteristics and accounting measures. We address the database as follows: First, following Brandt et al. (2009), we rematch the firm panel data with firm code, firm name, legal person name, telephone number, area code, main products, etc. Second, we delete some samples that contain missing important and abnormal indicators. Third, China's national economic classification standards changed in 2002 and 2011. We perform a unified conversion of industry codes before 2002, from *GB_T4754-1994* to *GB_T4754-2002*, and then change the standards after 2011, that is, from *GB_T4754-2011* to *GB_T4754-2002*. After the above code conversion, we obtained a comparable uniform industry code between 1998 and 2013, that is, the GB_T4754-2002 standard. Finally, the administrative codes of some counties have also changed. We converted the administrative codes of the counties into unified county-level codes in 2013.
5) *Innovation information*: Innovation data come from the *China Patent Database*, which covers all patent data applied for and published in the State Intellectual Property Office (SIPO) since the implementation of the "Patent Law of the People's Republic of China" in 1985. The data contain detailed information about each patent, such as the patent name, applicant, inventor, application date, applicant address, and a short description of the patent.

We used the firm name, address, and CEO name to merge the patent data with the firm survey data and then combined the meteorological data with the county's administrative code. Finally, we obtained a comprehensive database including firm information,

Office of Global Simulation and Assimilation of the NASA.

[6] Although there were no officially disclosed PM2.5 indicators before 2012, the public and enterprises could still understand the local air pollution levels through other visible pollutants and news reports.

innovation information, and meteorological data from 1998 to 2013. To avoid interference caused by outliers, we winsorized the main variables at 1%. The descriptive statistical results of the main variables are shown in Table 1.

5 Results

5.1 Baseline estimates

Table 2 reports the baseline results. The results in Columns (1)-(3) are from the 2SLS (two-stage least squares) estimation. The first column of Panel A in Table 2 does not add any control variables but only the fixed effect of the firm and year. In Columns 2–3, firm-level and weather variables are gradually added as control variables. Taking Column 3 as an example, the concentration of $PM_{2.5}$ has a significant negative impact on the number of patent applications. When the concentration of $PM_{2.5}$ increases by 1 μg/m^3, the number of patent applications significantly decreases by 0.72%. The first-stage estimation in Panel B shows that the instrument (the annual days of thermal inversion) has a positive and statistically significant effect on the concentration of $PM_{2.5}$. The Kleibergen–Paap Wald rk F-statistic for weak identification is much larger than the Stock-Yogo critical value of 16.38, rejecting the null hypothesis that our IV for air pollution is subject to the weak IV problem.

We further report estimates that do not account for any endogeneity bias. Column (4) in Table 2 presents the OLS estimates. This shows that the effect of $PM_{2.5}$ on firm innovation is positive and statistically significant. A positive effect of pollution on innovation could result from either simultaneity or omitted variable bias. For example, counties with more innovative firms may implement more advanced,

Table 1 Descriptive statistics

Variables	Definitions and Units	Mean	S.D.	Min	Max
Firm level:					
Patent applications	Log(patent applications + 1)	0.088	0.425	0	4.431
Firm size	Log(total assets)	9.982	1.482	6.937	14.30
Leverage	Total corporate liabilities/total assets	0.570	0.361	0	2.032
Firm age	Log(establish year-current year)	2.071	0.825	0	3.951
Capital density	Net fixed assets/total assets	0.362	0.237	0.004	0.942
Inventory density	Firm inventory/total assets	0.177	0.169	0	0.777
Profitability	Total profit/total assets	0.118	0.229	−0.217	1.219
Air pollution:					
$PM_{2.5}$ concentration	$PM_{2.5}$ concentration in a specific year(μg/m^3)	38.469	10.568	2.414	92.23
IV:					
Annual days	The annual number of days with thermal inversion	185.410	67.865	0	340

The sample period covers 1998–2013, and we have not given descriptive results for meteorological variables

Table 2 Baseline Estimation

Model	2SLS	2SLS	2SLS	OLS	Reduced-form
	(1)	(2)	(3)	(4)	(5)
Panel A. Second-stage estimation. Dependent variable:	Log Allpatent	Log Allpatent	Log Allpaten		
PM2.5 (instrumented)	−0.0035***	−0.0036***	−0.0072***		
	(0.0013)	(0.0013)	(0.0014)		
Panel B. First-stage estimation. Dependent variable:	$PM_{2.5}$	$PM_{2.5}$	$PM_{2.5}$		
Annual days	0.0142***	0.0141***	0.0134***		
	(0.0003)	(0.0003)	(0.0002)		
Kleibergen–Paap Wald F-statistic	2853.616	2942.865	2997.772		
Panel C. Reduced-form and OLS estimation. Dependent variable:				Log Allpatent	Log Allpatent
$PM_{2.5}$				0.0005***	
				(0.0001)	
Annual days					−0.0001***
					(0.0000)
size		0.0382***	0.0383***	0.0381***	0.0381***
		(0.0007)	(0.0007)	(0.0007)	(0.0007)
lev		−0.0040***	−0.0046***	−0.0034***	−0.0034***
		(0.0011)	(0.0011)	(0.0011)	(0.0011)
lnage		−0.0152***	−0.0155***	−0.0151***	−0.0151***
		(0.0007)	(0.0007)	(0.0007)	(0.0007)
capital		−0.0065***	−0.0064***	−0.0073***	−0.0073***
		(0.0017)	(0.0017)	(0.0017)	(0.0017)
invent		0.0204***	0.0200***	0.0203***	0.0204***
		(0.0021)	(0.0022)	(0.0021)	(0.0021)
roa		−0.0161***	−0.0146***	−0.0168***	−0.0169***
		(0.0014)	(0.0014)	(0.0013)	(0.0013)
Weather variables	No	No	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
N	3,259,180	3,201,409	3,193,284	3,193,284	3,193,284

The table reports the baseline estimation that examines the effect of air pollution on firm innovation. The data are annual from 1998 to 2013. The difference in sample observations is due to the absence of some control variables. Robust standard errors clustered at the firm level are reported in parentheses. * $p<0.1$.** $p<0.5$.*** $p<0.01$

lower-polluting technology over time, leading to an upward bias. In addition, higher economic activity may also lead to more innovation (inherent advantages for technology diffusion) and more pollution (industrial agglomeration), which also biases them upward toward or above zero. We also check the reduced-form estimate. Column (5) of Table 2 reveals that the estimated coefficient of the instrumental variable is negative and statistically significant.

5.2 Impact on the quality of innovation

In addition to quantity, we also care about the impact of air pollution on innovation quality. Generally, the quality of innovation is measured by patent citations. However, Hall et al. (2005) noted that the direct use of patent citations will face interference from interindustry heterogeneity and patent self-introduction. The type of patent is also a good indicator of the quality of the patent. There are three main types of patents in China: invention patents, utility model patents, and design patents. Cai et al. (2018) noted that the innovation quality of these three kinds of patents is gradually declining. Therefore, we use the logarithm of these three patents as the index to measure innovation quality. The results are shown in Columns (1)-(3) in Table 3. Although air pollution has a significant negative effect on different types of patents, it has a greater impact on invention patents, with a coefficient 2 to 3 times greater than that of the other two types of patents.

The IPC classification number format in the firm patent documents of China's National Intellectual Property Administration (CNIPA) is generally divided into "Section - Main Class - Subclass - Main Group - Subgroup", such as "A01B01/00". The first letter of the classification number represents eight major sections (Section A: Human Necessities (Agriculture, Light Industry, Medicine); Section B: Operations and Transport; Section C: Chemistry and Metallurgy; Section D: Textiles and Paper; Section E: Fixed Constructions; Section F: Mechanical Engineering, Lighting, Heating, Weapons, Blasting; Section G: Physics; Section H: Electrical Technology). The 2nd to 3rd digits indicates the main class, and the 4th character represents the subclass.

We use patent knowledge to reflect patent quality. First, simply measuring patent quality by the number of main classifications cannot accurately distinguish internal differences in patents, which may lead to overestimation of patent quality. For example, suppose firm A has three patents with main classification numbers A01B01/01,

Table 3 Impact on innovation quality

Model	2SLS	2SLS	2SLS	2SLS
	(1)	(2)	(3)	(4)
Panel A. Second-stage estimation. Dependent variable:	Log Invention	Log Utility	Log Design	Patent knowledge
$PM_{2.5}$ (instrumented)	−0.0058***	−0.0025**	−0.0026***	−0.0018***
	(0.0008)	(0.0011)	(0.0008)	(0.0005)
Panel B. First-stage estimation. Dependent variable:	$PM_{2.5}$	$PM_{2.5}$	$PM_{2.5}$	$PM_{2.5}$
Annual days	0.0134***	0.0134***	0.0134***	0.0134***
	(0.0002)	(0.0002)	(0.0002)	(0.0002)
Kleibergen–Paap Wald F-statistic	2997.772	2997.772	2997.772	2997.788
Time-varying firm controls	Yes	Yes	Yes	Yes
Weather variables	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
N	3,193,284	3,193,284	3,193,284	3,204,277

The table reports the IV estimates that examine the effect of air pollution on innovation quality. The data are annual from 1998 to 2013. The coefficients of the control variables are not reported. Robust standard errors clustered at the firm level are reported in parentheses.* $p<0.1$.** $p<0.5$.*** $p<0.01$

A01B01/02, and A01B01/03, while firm B has three patents with main classification numbers A01B02/00, B02C03/00, and C03D04/00. Although both have the same number of patent classification numbers, firm A's patents only utilize information from one main group (A01B01), whereas firm B's patents cover three main groups (A01B02, B02C03, C03D04). Obviously, the knowledge breadth of firm B's patents is greater than that of firm A, indicating higher patent quality.

Drawing on the calculation logic of the Herfindahl Index, we define the patent knowledge at the large group level as:

$$Patentknowledge_{it} = 1 - \sum \left(\frac{Z_{imt}}{Z_{it}} \right)^2$$

where the subscripts i, m, and t represent the firm, large group, and year, respectively. Z_{imt} is defined as the cumulative number of patent applications of firm i in Group m up to year t. Z_{it} is the cumulative number of patent applications under all major groups of firm i up to year t. The greater the patent knowledge is, the greater the difference between the patent classification numbers at each large group level and the higher the patent quality. The results are shown in Column (4) in Table 3, where we can see that air pollution has a significant negative impact on patent knowledge.

5.3 Consider environmental regulation

From the previous analysis, we found that the impact of air pollution on firm innovation is significantly negative. However, there is a possibility that environmental regulation may affect both pollution levels and firm innovation. For example, if a county has stringent emission standards, the air pollution level might be low, and the firms may respond to the strict standards by more patenting (Popp 2004). If this happens, it will have a severe impact on our conclusions. Therefore, in this section, we consider the impact of environmental regulation on our empirical research. However, since environmental regulation is a subjective indicator that is difficult to quantify directly, we start from other policies that may affect the intensity of environmental regulation and look for appropriate proxy variables.

First, we use the "Acid Rain Control Zone and Sulfur Dioxide Pollution Control Zone Division Plan" implemented by China in 1998 as a proxy indicator to measure the strength of regional environmental regulation. This plan puts sulfur dioxide and acid rain as key control objects and divides 175 cities within the "Two Control Zone" (TCZ). These cities will implement more stringent environmental control policies[7] (Chen et al. 2018a, b). Therefore, we add the interaction term of whether the firm is within the scope of the "TCZ" with the year dummy. The results are shown in Column (1) of Table 4, where we can see that air pollution still has a significant negative impact on firm innovation. Considering environmental regulations does not significantly change the basic conclusion.

Second, we measure the intensity of regulation from local government investment in environmental governance. The logic of this measurement is that the greater the

[7] The pollution control measures in the "TCZ" include prohibiting new coal mines with a sulfur content of 3% and above, and coal mines that already use coal of similar quality must gradually reduce or suspend production.

Table 4 Environmental regulation

Model	2SLS	2SLS	2SLS
	(1)	(2)	(3)
Panel A. Second-stage estimation. Dependent variable:	Log Allpatent	Log Allpatent	Log Allpatent
$PM_{2.5}$ (instrumented)	−0.0063***	−0.0085***	−0.0076***
	(0.0014)	(0.0017)	(0.0017)
Panel B. First-stage estimation. Dependent variable:	$PM_{2.5}$	$PM_{2.5}$	$PM_{2.5}$
Annual days	0.0136***	0.0117***	0.0117***
	(0.0002)	(0.0002)	(0.0002)
Kleibergen–Paap Wald F-statistic	3728.624	3759.862	3750.465
Time-varying firm controls	Yes	Yes	Yes
Weather variables	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
TCZ × year	Yes		
Investment in pollution source treatment × year		Yes	
Investment in urban environmental facilities × year			Yes
N	3,193,284	3,118,539	3,115,574

Data are annual from 1998 to 2013 at the county level. The coefficients of the control variables are not reported. Robust standard errors clustered at the firm level are reported in parentheses.* $p<0.1$. ** $p<0.5$.*** $p<0.01$

local government's investment in pollution control is, the greater the importance of environmental quality within its jurisdiction and the greater the likelihood that it will improve environmental regulation. Specifically, we add the interaction terms of the initial investment level in pollution source treatment and urban environmental facilities with the year dummy[8]. The results are shown in Columns (2)-(3) of Table 4, where the basic conclusion remains unchanged.

5.4 Robustness checks

Table 5 reports a series of robustness checks, in which Panel A reports the second-stage regression and Panel B reports the first-stage regression. Panel B shows that the instrumental variables have a significant positive effect on the endogenous variables, and the Kleibergen–Paap F statistics are much larger than the Stock-Yogo threshold. The results of the second-stage regression are analyzed below.

1) *Change the pollution variable.* In the benchmark regression, we use the $PM_{2.5}$ concentration to measure the county's air pollution level, but in addition to $PM_{2.5}$, sulfur dioxide (SO_2) is also a common pollutant. Therefore, we use the county's SO_2 concentration to represent air pollution. According to the results in Column 1 of Table 5, the concentration of SO_2 still has a significant negative impact on firm innovation. The basic conclusion remains unchanged.

[8] The "*China City Statistical Yearbook*" began publishing information on urban environmental governance investment in 2004, so the initial investment in urban environmental governance refers to the level of investment in 2004.

Table 5 Robustness checks

Model	2SLS	2SLS	2SLS	2SLS	2SLS	2SLS	2SRI
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A. Second-stage estimation. Dependent variable:	Log Allpatent	Patent Intensity	Log Allpatent	Log Allpatent	Log Allpatent	Log Allpatent	Log Allpatent
SO_2(instrumented)	−0.1415***						
	(0.0280)						
$PM_{2.5}$ (instrumented)		−0.0004*	−0.0132***	−0.0075**	−0.0062***	−0.0038***	
		(0.0002)	(0.0023)	(0.0014)	(0.0014)	(0.0010)	
$PM_{2.5}$ (2SRI)							−0.2453***
							(0.1074)
Panel B. First-stage estimation. Dependent variable:	SO_2	$PM_{2.5}$	$PM_{2.5}$	$PM_{2.5}$	$PM_{2.5}$	$PM_{2.5}$	$PM_{2.5}$
Annual days	0. 0007***	0.0133***	0.0089***	0. 0116***	0.0134***	0.0187***	0. 0023 ***
	(0.0000)	(0.0002)	(0.0001)	(0.0016)	(0.0002)	(0.0002)	(0.0001)
Kleibergen–Paap Wald F-statistic	1089.749	3068.038	5740.780	53.019	5944.154	1.4e+04	2415.51
Time-varying firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Weather variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province-specific quadratic time trends						Yes	
N	3,193,284	3,167,513	2,507,020	3,191,962	2,865,822	3,193,284	3,348,243

The table reports the robustness checks. The data are annual from 1998 to 2013. Colum 7 use Poisson regression model.* $p<0.1$.** $p<0.5$.*** $p<0.01$

2) *Change innovation variable.* Considering the significant differences in the number of patent applications by firms of different sizes, we use the intensity of patent application to measure the level of firm innovation, defined as the number of patent applications divided by the total number of employees of the firm. The results are shown in Column 2 in Table 5, where we can see that air pollution is also not conducive to the intensity of patent applications.
3) *Consider Firm sorting*. A potential threat to the baseline conclusion is firm sorting. For example, firms may respond to poor air quality by relocating to places with cleaner air. If there is a correlation between the characteristics of these firms and air pollution, the estimates might be biased. Therefore, in Column 3 of Table 5, we exclude firms that relocated to different counties during the sample period (approximately 10% of the whole sample). The coefficient of air pollution on firm innovation is slightly larger than that of the baseline estimation, but it is still statistically significant.
4) *Change in cluster pattern.* In the baseline estimation, we cluster the SEs at the firm level; however, there may be time-invariant unobserved factors affecting innovation at the county level. In Column 4 of Table 5, we cluster the SEs at the county level,[9] which allows for spatial and serial correlation within the county. It can be seen that changing cluster patterns will not impact the empirical results.
5) *Restricted sample.* In the baseline regression, the sample of firms is distributed across all regions of the country. However, there may be some systemic differences in innovation levels between firms located in the affiliated counties of provincial capital cities and firms in other counties. Therefore, in Column 5 of Table 5, we exclude the firms located in the affiliated counties of provincial capital cities. The coefficient of air pollution on firm innovation has not changed much, and it is still significant at the 1% level.
6) *Regional trends.* Our baseline results use year fixed effects to control for nationwide time trends. However, for different regions, there may be different development trends. Therefore, in Column 6 of Table 5, we use year fixed effects along with province-specific quadratic time trends to further reduce endogeneity bias. After adding the province-specific trend, the basic conclusion remains unchanged.
7) *Poisson regression.* Since the number of patent applications is a non-negative integer, this contradicts the assumption of traditional OLS regression that the dependent variable is continuous and normally distributed. Therefore, we used the Poisson regression to conduct a robustness test (Cohn et al. 2010). It is worth noting that our core explanatory variable has an endogeneity problem. Therefore, following on Terza et al. (2008), we used the two-stage residual inclusion method (2SRI), which is an extension of the two-stage least squares estimation (2SLS) and is widely used to solve the endogeneity problem of nonlinear models. The specific operation is to regress the endogenous variable on the instrumental variable and all exogenous variables in the first stage, add the residuals from the first stage in the second stage regression, and then regress the dependent variable on the fitted values of the residuals. The results of the instrumental variable regression show that air pollution has a significant negative impact on the number of firm patent applications, indicating that the baseline results remain robust.

[9] We also try to cluster the SEs at county-year level. The results remain unchanged.

5.5 Heterogeneity analysis

First, according to the type of firm registration, we divided the sample into state-owned firms, private firms, and foreign firms (including firms invested in Hong Kong, Macao, Taiwan, and foreign-funded firms). According to Columns (1)-(3) in Table 6, the negative impact of air pollution on firm innovation is more reflected in private firms, and the influence on state-owned firms and foreign firms is relatively low and insignificant. The reason may be that compared with private firms, state-owned firms have lower labor loss, and the overall employment environment is relatively stable. Therefore, marginal efficiency loss may be low in the face of the negative impact of air pollution. For foreign firms, the management and technology levels are much higher. Thus, the impact of air pollution is still relatively low. In addition, unlike those of private firms, the welfare systems of state-owned firms and foreign firms are complete, and they can improve their self-improvement in the face of the damage to labor productivity caused by air pollution and reduce the impact on innovation.

Second, we divided firms by the capital-labor ratio, which is defined as the ratio of total assets to employees. Then, the firms with a capital-labor ratio greater than the 75% quantile level are defined as capital-intensive firms. In comparison, those with a capital-labor ratio less than the 25% quantile level are defined as labor-intensive firms. Columns (4)-(5) in Table 6 show that the concentration of $PM_{2.5}$ has a significant and negative impact on capital-intensive firms but has no significant impact on labor-intensive firms. One possible explanation is that low-skilled labor accounts for a relatively large proportion of labor-intensive companies. They are less sensitive to air quality, and labor loss is not severe. For capital-intensive companies, the high-skilled labor force is relatively large. When facing

Table 6 Heterogeneity analysis

Model	2SLS	2SLS	2SLS	2SLS	2SLS
	(1) State-owned firms	(2) Private firms	(3) Foreign firms	(6) Capital-intensity firms	(7) Labor-intensity firms
Panel A. Second-stage estimation. Dependent variable:	Log Allpatent	Log Allpatent	Log Allpatent	Log Allpatent	Log Allpaten
$PM_{2.5}$ (instrumented)	−0.0044	−0.0068***	0.0053	−0.0083**	0.0010
	(0.0039)	(0.0014)	(0.0054)	(0.0036)	(0.0015)
Panel B. First-stage estimation. Dependent variable:	$PM_{2.5}$	$PM_{2.5}$	$PM_{2.5}$	$PM_{2.5}$	$PM_{2.5}$
Annual days	0. 0160***	0.0145***	0. 0099***	0.0157***	0.0141***
	(0.0009)	(0.0003)	(0.0004)	(0.0005)	(0.0005)
Kleibergen–Paap Wald F-statistic	449.677	2370.357	691.848	1080.462	918.559
Time-varying firm controls	Yes	Yes	Yes	Yes	Yes
Weather variables	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
N	275,280	2,296,122	598,617	735,467	698,807

The table reports the heterogeneity analysis. The data are annual from 1998 to 2013. The coefficients of the control variables are not reported.* $p<0.1$.** $p<0.5$.*** $p<0.01$

harmful air pollution, the loss of human capital is profound, so air quality has a significant impact on patent applications.

5.6 Mechanism

Air pollution harms health and therefore is not conducive to the accumulation of human capital (Brunekreef and Holgate 2002; Künzli et al. 2005; Chen et al. 2013; Ebenstein et al. 2015). In highly polluted areas, high-skilled workers tend to migrate outward, affecting the level of firm innovation. However, except in 2004, the *Annual Surveys of Manufacturing Firms* did not report the types of employees, and data related to high-skilled human capital at the firm level could not be obtained. Therefore, in testing the mechanism of human capital accumulation, we mainly use the *National Innovation Survey Firm Database* published by the National Bureau of Statistics to report the indicators of manufacturing firms engaged in technological activities, especially to provide detailed information about R&D workers. We use the firm code to match the annual surveys of manufacturing firms and obtain the database range from 2008 to 2013.[10]

We constructed two indicators to measure the level of human capital in each firm. One is to use the proportion of R&D workers in the total number of employees, and the other is the proportion of employees who have obtained bachelor's, master's, or doctoral degrees in the total number of employees. The results are shown in Columns 1 and 2 in Table 7. The concentration of $PM_{2.5}$ has a significant negative effect on firm human capital [11].

In addition, poor air quality may also increase production costs, squeeze out cash flow, and be detrimental to a firm's R&D investment. When air quality is poor, on the one hand, expenditures on environmental protection, such as the need to purchase more pollution treatment equipment, may increase significantly. On the other hand, firms will also improve the welfare system and purchase more air purification devices to alleviate the negative impact of air quality on the labor force. As shown in Column 3 in Table 7, air pollution will significantly increase the production costs of firms. Specifically, as the concentration of $PM_{2.5}$ increases by 1 μg/m^3, the production cost significantly increases by 3.08%.

To further verify this point of view, we use the pollution discharge fees disclosed by the *Annual Surveys of Manufacturing Firms* as the explained variable[12]. The results are shown in Column (4) of Table 7. The concentration of $PM_{2.5}$ has a significant positive impact on a firm's pollutant discharge fees. When the concentration of $PM_{2.5}$ increases by 1 μg/m^3, the discharge fee significantly increases by 1.92%.

[10] The National Bureau of Statistics has published only the *National Innovation Survey Firm Database* for 2008–2014.

[11] There may also be a situation which the R&D department of the firms and the production department are not in the same place. Unfortunately, there is no such information in the *Annual Surveys of Manufacturing Firms*, and we are unable to verify this result. However, we believe that the empirical conclusion based on the fact that the place where the R&D worker is located is consistent with the production place is actually the lower limit of the impact of air pollution on human capital. Considering that high-skilled labor prefers to work in cleaner areas, the air pollution will have a more significant impact on human capital.

[12] It should be noted that the *Annual Surveys of Manufacturing Firms* only reported firm-level pollution discharge fees in 2004, so we use cross-sectional data for analysis.

Table 7 Mechanisms

Model	2SLS	2SLS	2SLS	2SLS
	(1)	(2)	(3)	(4)
Panel A. Second-stage estimation. Dependent variable:	The proportion of R&D workers in the total employees	The proportion of employees with a bachelor's degree or above	Log production cost	Log discharge fees
$PM_{2.5}$ (instrumented)	−0.0075**	−0.0021*	0.0308***	0.0192***
	(0.0032)	(0.0011)	(0.0021)	(0.0008)
Panel B. First-stage estimation. Dependent variable:	$PM_{2.5}$	$PM_{2.5}$	PM2.5	PM2.5
Annual days	0.0176***	0.0176***	0.0180***	0.0797***
	(0.0005)	(0.0005)	(0.0003)	(0.0003)
Kleibergen–Paap Wald F-statistic	1114.894	1114.894	4104.23	7.5e+04
Time-varying firm controls	Yes	Yes	Yes	Yes
Weather variables	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	
Year fixed effects	Yes	Yes	Yes	
N	196,724	196,724	2,136,391	245,580

The table reports the effects of air pollution on firm innovation. The data are annual from 1998 to 2013. The dependent variable in Columns 1–2 is the proportion of R&D workers among the total number of employees and the proportion of employees who have obtained bachelor's, master's, or doctoral degrees among the total number of employees. The dependent variable in Columns 3–4 is the logarithm of production cost and the logarithm of firm discharge fees, respectively. The coefficients of the control variables are not reported. Robust standard errors clustered at the firm level are reported in parentheses.* $p<0.1$.** $p<0.5$.*** $p<0.01$

Combined with the previous analysis, when the air quality is poor, on the one hand, high-skilled workers "vote with feet" to choose to flow out; on the other hand, with the increase in production costs, firms may further squeeze R&D investment.

5.7 Further analysis

In addition to affecting firm innovation, does air pollution also impact other firm behaviors? In this section, we will conduct a further analysis of the impact of air pollution on firm performance. We use the firm financial information provided in the *Annual Surveys of Manufacturing Firms (ASMF)* to conduct empirical tests. Specifically, we mainly use the growth rate of firm sales and the growth rate of return on investment to measure firm performance. For the former, we primarily use the annual growth rate of firm sales for measurement. For the latter, we first standardize operating profits by total assets and then calculate their annual growth rate. The specific empirical results are shown in Table 8 below. It can be seen that air pollution also has a significant negative impact on firm performance.

6 Conclusion

This study estimates the causal effect of air pollution on firm innovation. To address endogeneity issues, we take an instrumental variable approach. We found that air pollution significantly reduces firm innovation. Specifically, as the concentration of

Table 8 Impact on firm performance

Model	2SLS	2SLS
	(1)	(2)
Panel A. Second-stage estimation. Dependent variable:	Growth rate of sales	Growth rate of return on investment
$PM_{2.5}$ (instrumented)	−0.0692***	−0.0248***
	(0.0126)	(0.0072)
Panel B. First-stage estimation. Dependent variable:	$PM_{2.5}$	$PM_{2.5}$
Annual days	0. 0042***	0. 0139***
	(0.0003)	(0.0002)
Kleibergen–Paap Wald F-statistic	186.085	3607.297
Time-varying firm controls	Yes	Yes
Weather variables	Yes	Yes
Firm fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
N	1,094,365	1,696,224

The table reports the estimates that examine the effect of air pollution on firm performance. The data are annual from 1998 to 2013. The coefficients of the control variables are not reported. Robust standard errors clustered at the firm level are reported in parentheses.* $p<0.1$.** $p<0.5$.*** $p<0.01$

$PM_{2.5}$ in the county increases by 1 μg/m3, the number of patent applications for the firm significantly decreases by 0.72%. The conclusion remains under a series of robustness checks. We further found that air pollution has a significant negative impact on the quality of innovation. In addition, air pollution has a heterogeneous impact on innovation. For private and capital-intensive firms, air pollution has a more significant impact on the number of patent applications. From the perspective of mechanism, we found that air pollution mainly affects innovation through channels that are not conducive to the accumulation of human capital and increase the production costs of firms. Further analysis has also found that air pollution is detrimental to corporate performance. The findings indicate that when assessing the economic and social impact of air pollution, its impact on firm behavior should not be ignored. Otherwise, the social cost of air pollution will be underestimated.

This article offers certain implications for environmental and innovation policy in developing countries. First, to encourage firm innovation, governments should not only implement more specific fiscal and monetary policies, strengthen property rights protection, and adopt other institutional measures but also enhance soft power—such as fostering a suitable environment. In particular, developing countries attach greater importance to environmental protection to provide a favorable external environment. Second, implementing human capital protection programs is crucial. As human capital serves as a vital driving force for a nation's long-term development, measures like providing air health subsidies to R&D personnel in high-tech firms can alleviate the impact of pollution on these professionals. Finally, we propose establishing firm innovation cost hedging mechanisms. This could involve implementing green R&D tax subsidy policies, offering additional deductions for firms' pollution control technologies, and exploring government-subsidized insurance premiums to cover R&D losses arising from production stoppages due to severe environmental pollution.

In addition, this study relies on the ASMF database, which covers only manufacturing firms with annual sales exceeding 5 million RMB (approximately $700,000 USD). While this threshold aligns with standard practices in national statistical systems, it systematically excludes micro-enterprises, small-scale innovators, and informal sector actors. Consequently, our findings on innovation drivers and productivity patterns may not generalize to these excluded segments of the economy.

Appendix

When studying air pollution in China, a strand of literature utilizes the policy of winter heating in northern China (north of the Qinling-Huaihe Line (Q-H)) to identify the impacts of air pollution through regression discontinuity design (Chen et al. 2013; Xue et al. 2021). Therefore, in this section, we adopt a similar identification strategy. Specifically, we specify the regression model as follows:

$$Y_{it} = \alpha + \beta_1 North_c + \beta_2 f(Lat_c) + \beta_3 North_c \times f(Lat_c) + \beta X_{it} + T_t + I_j + \epsilon_{it}$$

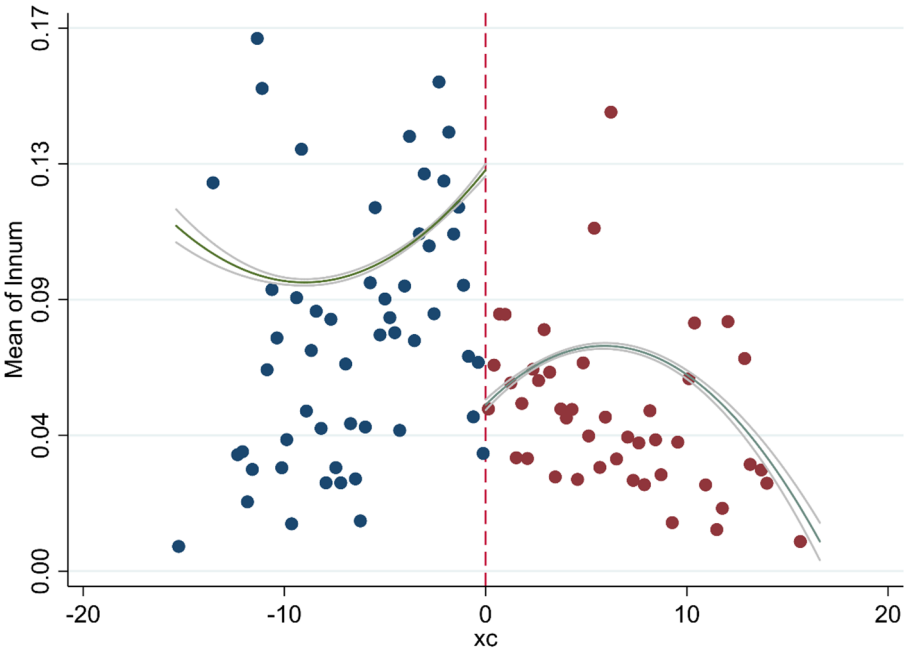

Fig. 3 Distribution of the average number of enterprise invention patents at the Qinling-Huaihe Boundary (The horizontal axis xc, representing the difference between the latitude of each city and the average latitude of the Q-H boundary. A value less than 0 indicates the southern region, while a value greater than 0 indicates the northern region. The vertical axis represents the mean of the logarithmic values of the number of patent applications)

Table A1 Results of regression discontinuity design

	(1)	(2)	(3)	(4)	(5)	(6)
	Log Allpatent	Log Allpatent	Log Allpaten	Log Allpaten	PM2.5	Log Allpaten
North	−0.0402***	−0.0461***	−0.0460***	−0.0351***	4.9007**	
	(0.0054)	(0.0081)	(0.0045)	(0.0102)	(2.1697)	
PM2.5						−0.0094**
						(0.004)
Kleibergen-Paap Wald rk F					12.901	
Time-varying firm controls	NO	NO	YES	YES	YES	YES
Weather variables	NO	NO	YES	YES	YES	YES
Industry fixed effects	YES	YES	YES	YES	YES	YES
Year fixed effects	YES	YES	YES	YES	YES	YES
polynomials	first-order	second-order	first-order	second-order	second-order	second-order
N	3,392,056	3,392,056	3,315,943	3,317,442	3,317,442	3,317,442

Note: Standard errors clustered at the city level are shown in parentheses. The differences in sample observations are due to missing values in some control variables. We control for interaction terms between treatment variables and polynomials of running variables.* $p<0.1$.** $p<0.5$.*** $p<0.01$

$North_c$ is the treatment variable, which takes the value of 1 if the city c is north of the Q-H boundary, and 0 otherwise. Lat_c is the running variable, we represent by the difference between the latitude of each city and the average latitude of the Q-H boundary. $f\left(Lat_c\right)$ denotes a smooth function of the running variable. In practice, we use different polynomials for testing, and we also control for interaction terms between the treatment variable and polynomials of the running variable. X_{it} can represent control variables at the firm levels, with specific definitions consistent with those in the baseline regression. We control for industry and year fixed effects. Additionally, we cluster the standard errors at city level.

Before conducting empirical analysis, we first observe whether there is a cut-off in the number of patent applications at both side of the Q-H boundary. As shown in Fig. 3, the average number of patent applications exhibits a significant cut-off at the Q-H boundary. The number of patent applications by firm located south of the boundary is significantly higher than that north of the boundary, indicating that air pollution has a significant negative impact on firm innovation.

We identify the impact of air pollution on firm innovation through a regression discontinuity (RD) strategy. The results are shown in Table A1. The first and second columns do not control for any covariates, but only use the first-order and second-order terms of the running variable polynomial, respectively. The third and fourth columns control for all covariates. It can be seen that identifying causal relationships through the RD strategy does not affect the empirical conclusions of the baseline regression.

To further test whether this cut-off operates through the mechanism of air pollution, we use two-stage least squares (2SLS) regression in the last two columns of Table A1, treating the treatment variable $North_c$ as an instrumental variable for air pollution. Column 5 shows the first-stage regression results, indicating a significant positive relationship between areas north of the Q-H boundary and PM2.5 concentration. The Kleibergen-Paap Wald rk F statistic passes the weak instrument test. Column 6 shows the second-stage results, revealing that using the Qi-H boundary as a spatial cut-off leads to conclusions similar to our baseline regression, i.e., air pollution significantly reduces the number of patent applications (Table A1).

Data availability The data that support the findings of this study are available from the corresponding author (pulong@zufe.edu.cn), upon reasonable request.

Declarations

Conflicts of interest None.

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